Draft version July 9, 2019
Typeset using LaTeX **twocolumn** style in AASTeX62

Pitch Angle Distribution Function of Local Spiral Galaxies and Its Role in Galactic Structure

Michael S. Fusco,[1,2] Benjamin L. Davis,[3] Daniel Kennefick,[1,2] Marc S. Seigar,[4] and Julia Kennefick[1,2]

[1] *Arkansas Center for Space and Planetary Sciences, Fayetteville, AR 72701*
[2] *Department of Physics, University of Arkansas, Fayetteville, AR 72701*
[3] *Centre for Astrophysics and Supercomputing, Swinburne University of Technology, Hawthorn, Victoria 3122, Australia*
[4] *College of Science and Engineering , University of Minnesota, Duluth MN, 55812*

ABSTRACT

We present an analysis of the Pitch Angle Distribution Function (PADF) and Black Hole Mass Function (BHMF) of a sample of nearby galaxies selected from the Carnegie-Irvine Galaxy Survey (CGS). Our subset consists of spiral galaxies with $M_B > -19.12$ and limiting luminosity distance $L_D < 25.4Mpc$ ($z = 0.00572$). These constraints result in a set of 74 spiral galaxies, 51 with measurable pitch angles. This is an extension of work by Davis et. al on a different subset of the CGS with the same limiting magnitude as our sample, but $\mathrm{M_b}$ brighter than -19.12, but with the same limiting magnitude as our sample. Their selection of galaxies was magnitude limited for purposes of sample completeness. Folding our subset into this study on local galaxies yields a group restricted only in luminosity-distance rather than both luminosity-distance and $\mathrm{M_b}$. We produce a BHMF for this combined sample, representative of spiral galaxies in the local universe.

The faint subset presented here is morphologically distinct from its brighter counterpart, with a higher percentage of Sc and Sd classified galaxies. However, we find the pitch angle distributions of the two samples are not strongly dissimilar. We explore some indications of populational differences as expressed through the distributions of galactic spiral arm pitch angles (and resulting BHMFs) of the two samples. These differences are especially expressed on the high pitch angle, or low Supermassive Black Hole (SMBH) mass tail of the distribution. Our data show a divide in the pitch angles of the a-c Hubble types and the cd,d,m types, with a-c types having tighter spiral arms on average. The cd,d, and m type galaxies therefore represent the lower mass end of spiral galaxies and form a mass distribution distinct from their a-c counterparts. Incorporation of the low mass end of late type galaxies into the BHMF is results in a more complete local BHMF. We present the behavior of this very late type, low mass regime of galaxies in the local universe.

Keywords: black hole physics, cosmology: miscellaneous, galaxies: evolution, galaxies: spiral

1. INTRODUCTION

Supermassive Black Holes (SMBHs) reside at the center of most galaxies (Richstone et al. 1998; Schödel et al. 2002). There is strong evidence supporting proposed relationships among the mass of the central SMBH and various properties and features of their host galaxies (for thorough review, see Graham et al. 2015, and references therein) and (Shu 2016). Given that black hole masses correspond to morphological features of their host galaxies, a comprehensive census can be undertaken to characterize the distribution of SMBH masses, A Black Hole Mass Function (BHMF) in the local universe. Extension of the BHMF to higher redshift galaxies would advance studies of galactic evolution through the time dependency of SMBH masses. The local census offers a snapshot of the current distribution of black hole masses in the present universe while a higher redshift study would

Corresponding author: Michael S. Fusco
msfusco@email.uark.edu

allow a mapping of the history of the BHMF through cosmic time. Incorporation of galaxies at greater redshifts will inform changes in the BHMF through processes such as mergers, accretion, and secular evolution.

At present, the high mass end of the BHMF has been probed through observations of the most active galaxies (Dalla Bontà et al. 2009). In the local universe, a majority of large black holes reside in elliptical or lenticular galaxies. The masses of these objects are estimated through a correlation between their central velocity dispersion, σ, and the mass of the SMBH (Ferrarese & Merritt 2000; Gebhardt et al. 2000). In general, the BHMF relies on applying scaling relations to luminosity/velocity functions, and has been utilized by many authors for populations of late-type and early-type galaxies (Salucci et al. 1999; Richstone 2002; Shankar et al. 2004; Shankar 2009; Tundo et al. 2007).

For the low mass-end of the BHMF, there is still no clear consensus as to the behavior of the distribution (Marconi et al. 2004; Shankar et al. 2004; Vika et al. 2009). A secure estimate of the low mass-end of the BHMF provides constraints on the degree of time evolution in the Eddington ratio distributions (e.g., Shankar 2009), as well as on the population of seed SMBHs (e.g., Natarajan 2011). To probe evolution of the BHMF over the lifetime of the universe, a more complete measurement of the distribution locally must be attained and modified to include galaxies of lower SMBH masses. Therefore, our exploration continues a census of black holes in late type galaxies, concentrating here on galaxies with lower absolute B-Band luminosities. As we will see, these galaxies represent the low mass tail of galactic black hole masses.

The most commonly used method of measurement for SMBH masses of early type galaxies is the M-σ relation between the mass of the black hole and the central stellar velocity dispersion of the galaxy. This relation has been extensively adopted to estimate the masses of early type galaxies in order to establish a BHMF for those morphologies (Franceschini et al. 1998; Graham et al. 2007). Another widely used mass measurement method for early type galaxies is the Sérsic Index (Savorgnan et al. (2013)). Spiral galaxies are structurally complex and require extensive bulge/disk decomposition before attempting to calculate either the central σ or an accurate Sérsic Index value. Neither of these methods have been convincingly shown to be effective in measuring SMBH masses for late type galaxies.

Here we use a correlation between the SMBH mass and the spiral galaxy pitch angle as proposed in Seigar et al. (2008) and Berrier et al. (2013), and developed further in Davis et al. (2015). According to the $M_{\rm BH}$–$P_{\rm spiral}$ relation, galaxies with larger SMBH masses will have more tightly wound spiral arms with lower pitch angles, whereas galaxies with lower SMBH masses will have looser spirals with higher pitch angles. This $M_{\rm BH}$–$P_{\rm spiral}$ relation has a few advantages over past techniques for measurement of SMBH masses in late type galaxies. The M–P relation has less a tighter correlation than other methods when applied to disk galaxies. More importantly, measuring the pitch angle of galaxy spiral arms only requires sufficiently resolved images of the galaxy rather than observationally intensive spectra. It may be, then, the best method for extending SMBH measurement in late type galaxies to higher redshifts is through the M–P relation rather than other current methods. As a result, the M–P relation is a useful method both for measuring the SMBH mass of spiral galaxies locally and at higher redshift, with the caveat that spiral structure be sufficiently resolved in imaging.

The M–P relation indirectly measures the mass of the central bulge region of spiral galaxies. Modal Density Wave Theory (Lin & Shu 1964) describes spiral structure in galaxies as a standing wave pattern of density waves. These density waves are generated by inner and outer Linblad resonance orbits in the galaxy. They propagate through the disk, amplified or damped by interactions with galactic features. The pitch angle of the spiral arms is analogous to the wavelength of a standing wave on a vibrating string. The value of pitch angle is dependent on the ratio of the mass density in the disk to the mass in the central region. The SMBH behaves like tension on the string, while the disk itself acts as the medium of the string. This formulation of modal density wave theory has been applied effectively to the behavior of spirals in the rings of Saturn (Cuzzi et al. 1981). Spiral galaxies, though more structurally complex than the high central mass,low disk mass ideal case, nevertheless have a configuration of a centrally concentrated mass (SMBH and bulge) with a thin disk of orbiting material. Since the mass of the galactic bulge correlates well with the SMBH mass, it follows that application of modal density wave theory to indirectly measure SMBH masses is a viable method. Unlike the end-member case of Saturn, where the disk mass is much less than the central mass, spiral galaxies have a comparatively flat mass distribution. Spiral structures induced by density waves are also present in protoplanetary and circumplanetary disks (Pérez et al. 2016; Xu & Goodman 2018; Bae et al. 2016). As the mass ratio of central mass to disk mass decreases, the logarithmic spiral arm pitch angle increases. While Saturn's ring spirals have a pitch angle much less than 1°, most spiral galaxies are closer to 20°. It has been shown in Davis et al. (2015) there is a planar re-

lationship among the quantities of SMBH mass, P, and the mass of neutral hydrogen (proxy for disk density) in the galaxy in accordance with Density Wave Theory.

A first census of the BHMF in spiral galaxies has been conducted by Davis et al. (2014), in which a sample of spiral galaxies was selected from the CGS (Ho et al. 2011) sample of nearby galaxies with two bounds, luminosity-distance and absolute B-Band magnitude. The reasoning for these bounds was for sample completeness, as fainter galaxies might not be identified beyond a specified distance. Here we consider these fainter galaxies in the larger sample, accounting for missing galaxies by weighted correction.

2. METHODOLOGY & DATA

2.1. *Sample Selection*

We make use of Southern Hemisphere galaxies based on the Carnegie-Irvine Galaxy Survey (CGS, Ho et al. 2011). The CGS is a project to observe 605 bright southern galaxies in photometry and spectroscopy using Las Campanas Observatory. For this research, we utilize optical (B-band) imaging of a subsample of these galaxies.

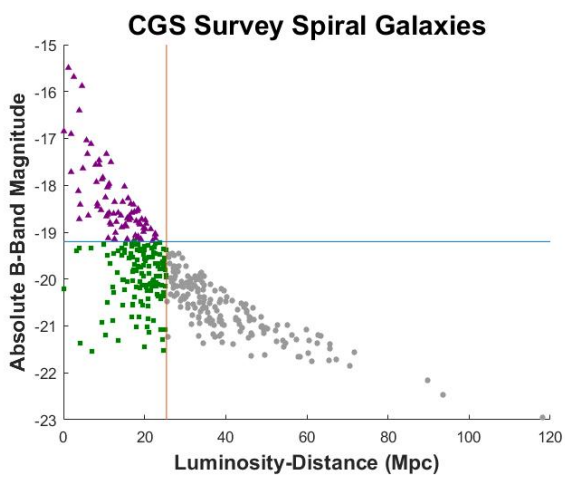


Figure 1. Luminosity Distance vs. Absolute B-Band magnitude for CGS Survey galaxies. Davis et al. (2014) sample (green squares); Faint galaxy sample (purple triangles).

Pursuit of a BHMF for the local universe is undertaken through the measurement of pitch angles to estimate black hole masses. The formerly defined 2014 sample provided an initial construction of the BHMF of late type galaxies, wherein the pitch angles of a volume-limited set of local spiral galaxies were measured to calculate the BHMF. Their dataset excluded galaxies below a specified luminosity (limiting magnitude $M_b = 19.12$); beyond the distance limit chosen, some fraction of these galaxies were beyond the detection limit of the survey. Consequently, their dataset is complete (no undetected galaxies) within its distance and luminosity bounds. Yet, these set limitations necessarily exclude the lowest luminosity galaxies, which may comprise the extreme low mass end of the BHMF. We measure pitch angles for all galaxies detected within the set luminosity-distance limit, accounting for those too faint to be identified. From here, we will refer to their sample as the bright sample and our sample as the faint sample.

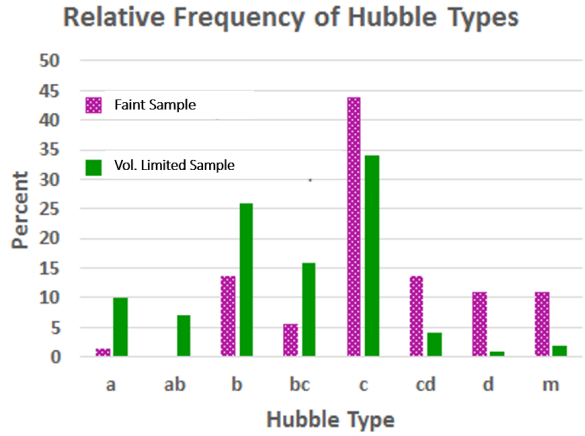
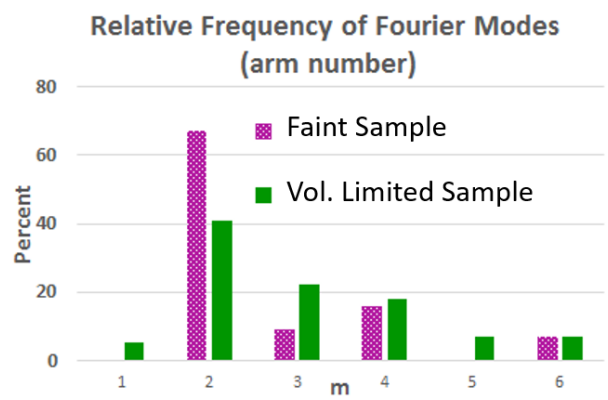


Figure 2. Above: Hubble types of all galaxies from each sample. The bright sample is comprised of spirals of medium arm winding (b, bc, c), while the faint sample favors less bulge dominated spirals with a high pitch angle sub-population (c, cd, d, m). Hence, a morphological difference is expressed between the two selections of galaxies. Below: Number of arms for the two samples is similar, with two armed galaxies being most abundant in both. Notably, the faint sample is partially comprised of flocculent galaxies which often evade pitch angle measurement or arm number attribution, as opposed to the more accurately measured grand design spirals which are common to the earlier Davis sample.



Our sample eliminates the limiting luminosity constraint to probe lower luminosity galaxies (see Figure 1). From this, we selected spiral galaxies within a set defined by a limiting luminosity (redshift-independent) distance, $D_L = 25.4$ Mpc, where the distances are found

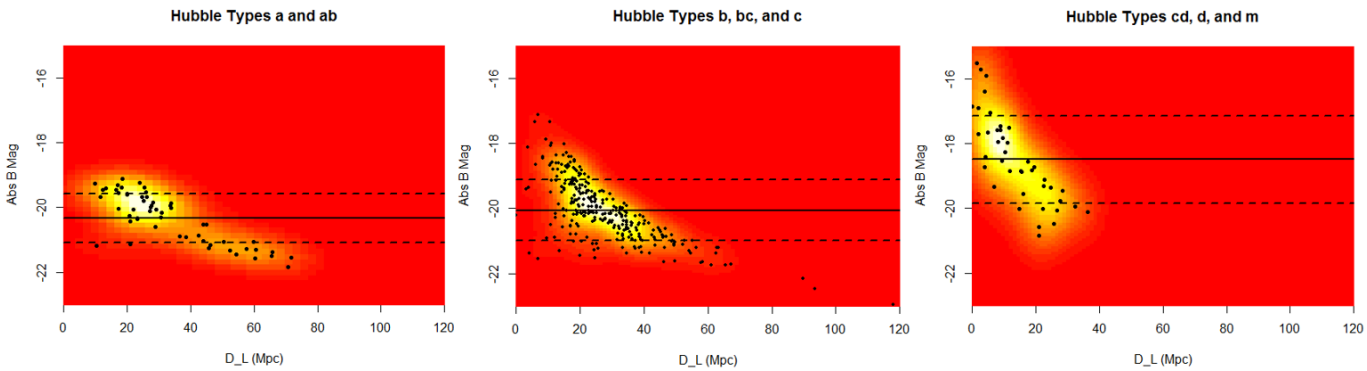


Figure 3. Mean absolute M_b and standard deviation of the full CGS sample, sorted by Hubble type. Hubble type a and ab spirals as reported in the CGS are confined to roughly $M_b = -19$ and brighter, with a mean near $M_b = -20.5$. Galaxies identified as b, bc, and c span nearly the whole sample in magnitude. Galaxies identified as c, cd, and m have a mean near $M_b = -18.5$. It is expected that galaxies with later Hubble types would have lower surface brightness, especially m dwarfs. Galaxy types through c in the CGS all have similar mean M_b, while the latest types begin to deviate toward lower brightness.

by primary techniques. This produces a volume-limited (but no longer magnitude limited) sample consisting of the 140 spiral galaxies of Davis et al. (2014) plus 74 faint galaxies in this study.

2.1.1. *Sample Comparison*

Here we consider the physical properties of the Davis sample compared to our fainter set. Two such properties are the Hubble type classification and modality (number of arms) of the galaxies (see Figure 2). Differences in these characteristics are indicators of distinct morphological populations of spiral galaxies comprising the two samples. In comparing the two sets of galaxies, the faint sample contains many more c and d type spirals, implying less bulge dominated and looser spiral armed galaxies on average. According to the Tully-Fisher relation (Tully & Fisher 1977) the luminosity of a spiral galaxy relates to its rotational velocity, which is determined in part by its central SMBH and bulge mass. The lower surface brightness galaxies of our sample are therefore likely to be less massive overall and less bulge dominated. The difference in Hubble classifications between the two samples confirms this expectation.

While the galaxies of the faint sample are different in Hubble type, their distribution of arm number mirrors that of Davis et al. (2014), where arms are distinguishable. Dominant symmetry in both sets of galaxies is $m = 2$ by far, with the faint sample showing an even higher percentage of two armed spirals. There is also an excess of flocculent galaxies in the faint sample. These generally fainter galaxies have a tendency toward less organized structure, with symmetry either unclear, having a low arm-interarm contrast, or being more difficult to identify with more spurs than grand design arms. Assigning a proper arm count to these messy galaxies is difficult. For fainter galaxies, it may simply be easier to resolve a two armed spiral than to count the number of arms in a less organized structure. This high number of flocculent galaxies explains the relative excess of $m = 2$ and missing $m > 2$ modes in the faint sample. As flocculent galaxies do not often have a dominant symmetry mode assigned to them, these missing assignments at least partially accounts for the dearth of galaxies with higher symmetry modes than $m = 2$ and corresponding excess of $m = 2$ galaxies in the faint sample.

2.2. *Galactic Spiral Arm Pitch Angle Measurement*

2.2.1. 2DFFT

Measurement of galactic logarithmic spiral arm pitch angle is achieved through utilization of a modified two dimensional fast Fourier transform code called 2DFFT (Davis et al. 2012, 2016; Puerari et al. 2000). This package decomposes digital images of spiral galaxies into superpositions of spirals of various pitch angles (P) and number of arms, or modality (m). Measurement using 2DFFT is most effective when applied to face-on galaxies. Galaxies at higher inclination to line of sight must be deprojected to a face-on orientation in pre-processing. Deprojection position and inclination angle are measured with the IRAF routine ELLIPSE. The intrinsic ellipticity of spiral galaxies is generally nonzero, though small, with an average of 0.215 ± 0.013 (Rodríguez & Padilla 2013). We here assume circular galaxies for our deprojections. Small errors in deprojection ($< 10°$ of inclination) do not greatly alter the measurement of pitch angle (Davis et al. 2012). Deviations from circularity of disks in these galaxies may increase the error value for pitch angle measurements.

Within the 2DFFT routine, the center of the galaxy and an outer radius are specified by the user, and pitch angles are measured in annuli with an inner and outer

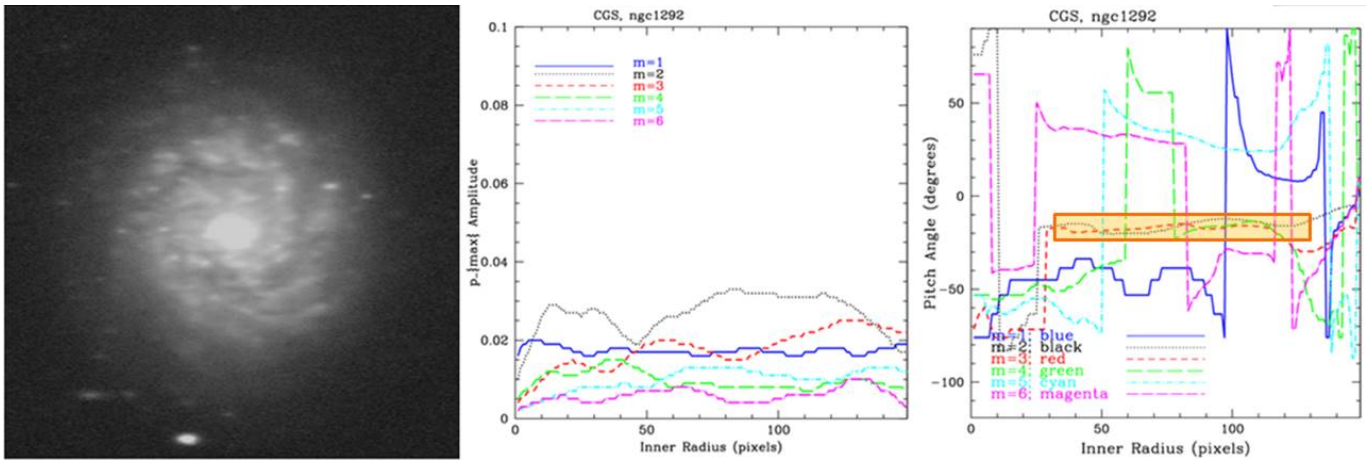


Figure 4. Example of spiral arm pitch angle measurement for galaxy NGC 1292 using 2DFFT. Left: Galaxy image in B-Band. Middle: Amplitude of each Fourier mode, with $m = 2$ symmetry being the most prominent for this galaxy. Right: 2DFFT results, with region of stable pitch angle measurement highlighted. Notice the stable region for the m=2 mode also shows stability for some less dominant, higher orders of symmetry, in this case $m = 3$ and $m = 4$. The two arm symmetry is distinguishable in the image, though this flocculent galaxy blends the $m = 2$ spiral with arm segments in different symmetry configurations.

radius. The inner radius is iterated from zero by integer pixel radius values to one less than the outer radius of the galaxy. This produces a set of measurements all with the same outer radius but of varying inner radius. Often, the largest annuli will contain the core of the galaxy and any bar features present, resulting in a poor measurement for pitch angle. An inner radius of measurement, outside of which the pitch angle is relatively stable (ideally a large fraction of the total galactic radius) is selected (see Figure 4). On the other end of the measurement, the smallest annuli are in the lowest surface brightness outer regions of the galaxy where signal to noise ratios are lower. These annuli also contain less of the rotation of the spiral as the annulus shrinks. These small outer annuli with unstable pitch angles are excluded. As a result, the measurements with small inner radii or large inner radii compared to the galaxy radius are excluded while a region of medium annuli with stable pitch angle measurements is used to find the dominant pitch angle in the galaxy. This stable region should be characterized by a small variation in the measured value for pitch angle. The error in pitch angle is dependent on the standard deviation of the pitch angle within the stable region as well as the size of the stable region compared to the size of the galaxy. 2DFFT also returns the amplitude of each of the first six nonzero harmonic modes. The dominant mode (number of arms) corresponds to the frequency with maximum amplitude. Full details of our methodology for measuring pitch angles may be found in (Davis et al. 2012).

2.2.2. *Use of* GALFIT *in Galaxies with low Arm-Interarm Contrast*

GALFIT version 3.0 is a two dimensional fitting algorithm for galaxies as described in (Peng et al. 2010). In addition to elliptical light profile fitting (Sérsic profiles, etc.), the up-to-date version of GALFIT allows detailed fitting of complex and non-axisymmetric features in galaxies such as spiral arms, bars, rings, tidal tails, etc. This has proven to be a valuable resource in addressing the less structured spiral galaxies.

There are two advantageous approaches in utilizing GALFIT to assist in pitch angle measurement for a given galaxy. The first is to fully model the galaxy in question, including spiral structure, and extract the pitch angle of the spirals from the model. The main advantage in this case is foreground stars and clumps in the galaxy structure go unmodeled and instead the focus is on reproducing radial light profiles and measurable spiral arms. A disadvantage to model fitting of spiral arms with GALFIT is the time cost per galaxy to fit a reasonable model. Spiral arms produced through GALFIT modeling behave a bit differently from real arms and may be difficult to match to real galaxy features. When working with large samples of galaxies, basic radial fitting without non-axisymmetric components is the preferred usage, and is the method employed for all galaxies in this sample.

The second possibility for incorporating GALFIT models into pitch angle measurement is to remove the radial light profile of the galaxy from the image before performing 2DFFT. This option offers a more practical route than full modeling. This technique is rapid to implement, and as a consequence is not prohibitively time intensive for measuring large samples of galaxies. Generally, a one or two component Sérsic profile (Sérsic

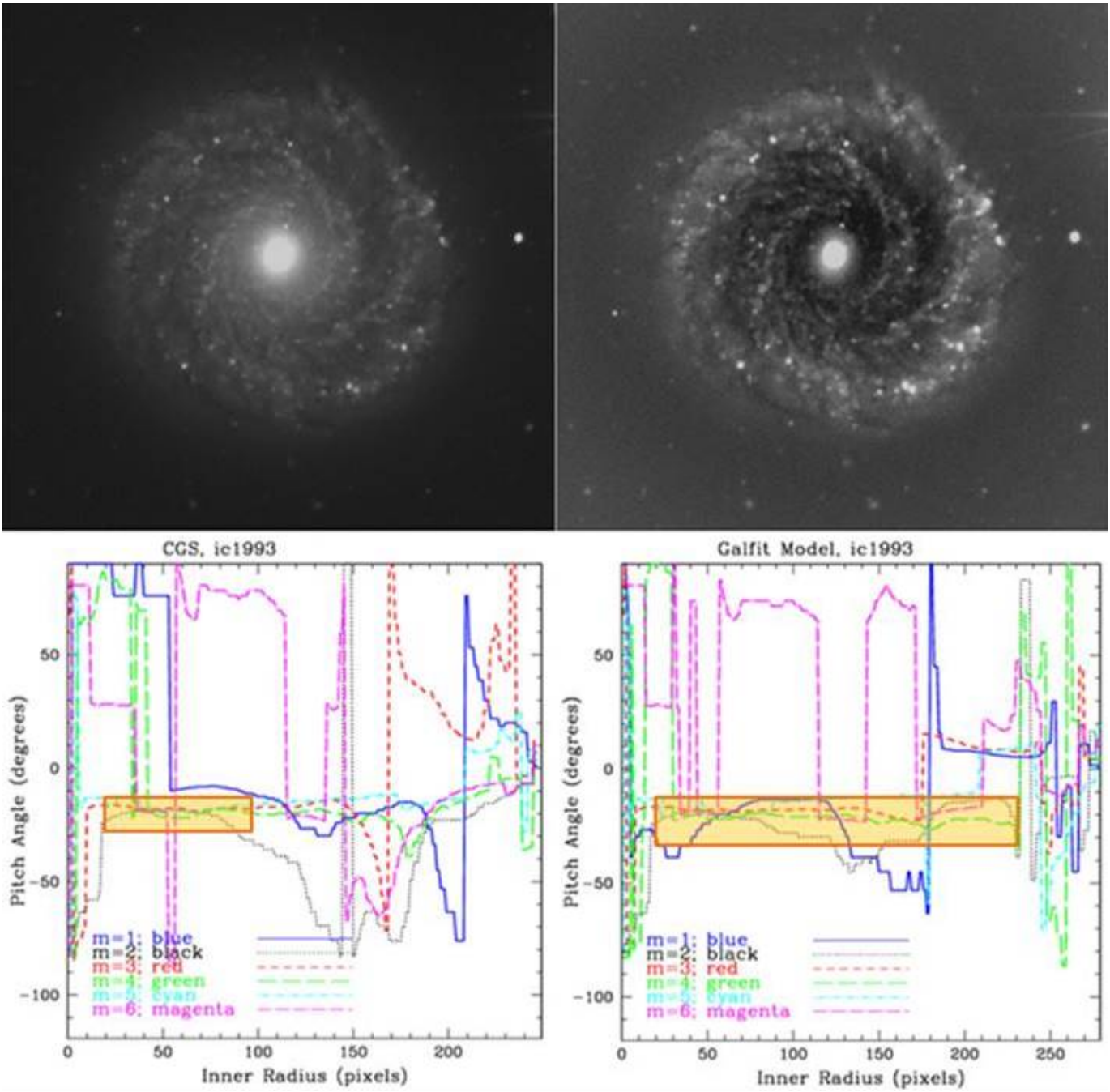


Figure 5. Improved 2DFFT measurement of spiral arm pitch angle for IC 1993 by means of GALFIT. Left: Original B-Band image and corresponding pitch angle measurement. Right: Modified image using GALFIT to remove a Sérsic profile from the galaxy, amplifying the visibility of the spiral arms. 2DFFT is capable of pitch angle measurement for both images; the stable region for the GALFIT corrected image covers a larger range of galactic radius with a smaller standard deviation, finding a lower measurement error.

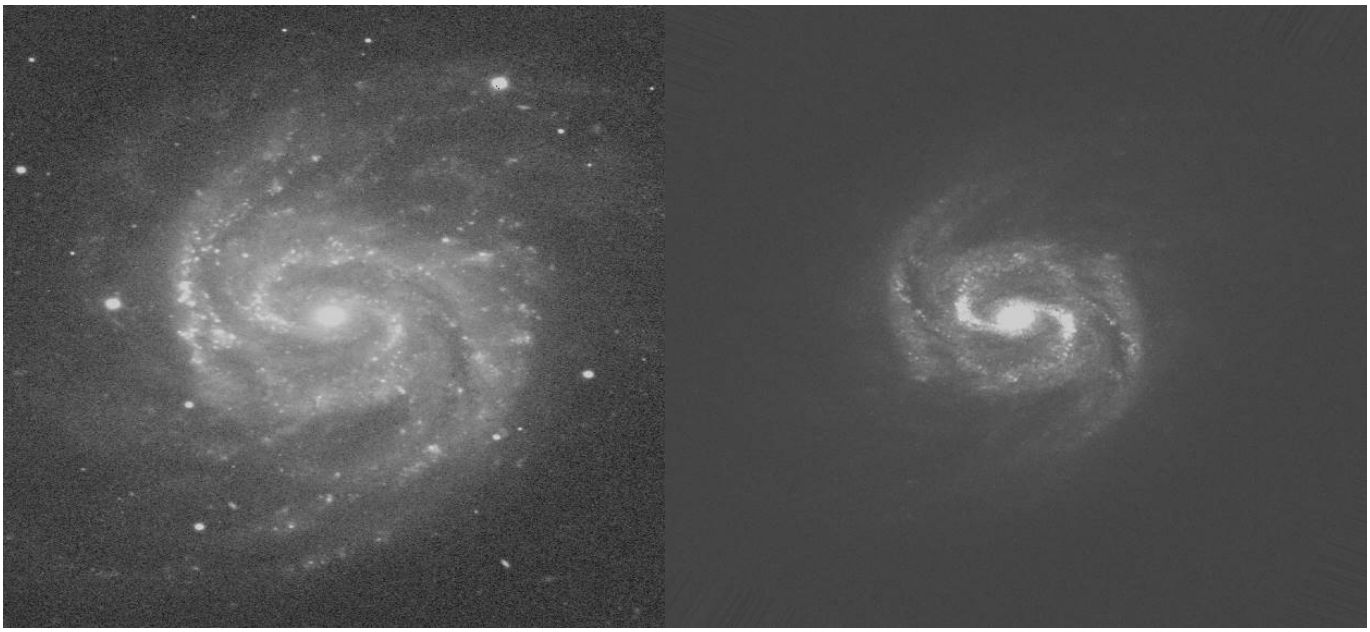


Figure 6. $m = 2$ symmetrical component of galaxy NGC 1042. Left: Original B-Band image. Right: Symmetrical component image. Foreground stars and asymmetric bright features are removed and symmetrical features remain, allowing for more accurate measurement of spiral arm pitch angles. Taking symmetrical components, where applicable, is a more rapid process for improving measurement than other methods, such as star subtraction. However, this comes at a cost of loss of information about the galaxy, as arm segments which do not fit the symmetry of the symmetrical component taken are removed.

1963) is fit to the target galaxy and removed, where the residual image primarily retains the spiral structure. Fitting the galaxy to an elliptical Sérsic profile is especially helpful in increasing the relative strength of spiral features at outer radii, as much of the central brightness of the galaxy is removed for fitting. Galaxies before GALFIT light profile removal may be considered as "center-weighted" as most of the brightness of the galaxy is centrally located. By flattening the radial light profile of the galaxy with GALFIT, subsequent pitch angle measurements with 2DFFT tend to be more stable and reach higher galactic radii (see Figure 5) as opposed to favoring the inner radius of a given annulus. By employing Sérsic profile subtraction, we obtain more and better pitch angle measurements than would be possible otherwise.

2.2.3. *Symmetrical Components*

Another technique for enhancing the accuracy of pitch angle measurements is to extract the symmetrical components of the target galaxy. Such a symmetrical component image is created by dividing the original image into pixel pairs/groups, co-radial from the center of the galaxy and symmetrically distributed in rotation angle (Elmegreen et al. 1992). Pairs/groups then adopt the lowest pixel value among them as the value for all of them. For example, for $m = 2$ symmetry, all points radially opposed from one another at 180° adopt the lowest pixel value of each pair; for $m = 4$, the lowest pixel value of the groups of four pixels 90° apart, and so on.

This method of image preprocessing is applicable to galaxies where the arm symmetry is apparent, but where foreground stars or structural irregularities cause inaccurate measurements of the overall behavior. Taking symmetrical components provides a trade with tension between two characteristics: effective resolution and structural irregularities. They tend to not only remove or reduce the effects of error inducing features such as foreground stars, minor arm spurs, and flocculence from the image (see Figure 6), but also remove detail from the galaxy. The effective pixel count of the galaxy is reduced by approximately a factor of the symmetry number as information is lost through taking the symmetry. In most cases, this is a worthwhile trade-off of lost resolution for improved measurement of symmetrical features with lower associated error. For galaxies with many foreground stars, a full star subtraction using DAOPHOT in IRAF or similar may be advisable before taking symmetrical component images for final measurement. Both GALFIT model subtractions and symmetrical component images may be introduced individually or in sequence to galaxy images. Thusfar, the most stable pitch angle measurements are obtained by first removing the radial Sérsic profile and subsequently creating a symmetrical component image of the residual from the Sérsic profile

fit. Applying these techniques in tandem often increases the measurable radius range for the galaxy by 10-25% of the galactic radius and reduces the error in pitch angle measurement for the galaxy by an average of roughly 2 degrees in our error assignment.

2.3. *Data*

Our sample of galaxies draws 74 spiral galaxies from the CGS, with absolute M_b fainter than -19.12 and limiting luminosity distance less than 25.4 Mpc. Of these, galactic spiral arm pitch angles were measured for 51 galaxies. Of the 23 galaxies which we were unable to measure, the Hubble type of the galaxy being examined greatly effected the measurement success rate. For a, b, bc, and c galaxies, the average measurement rate was 85%, whereas for cd and d galaxies the measurement rate drops to 67%. The measurement rate of Magellanic Spiral (m-type) dwarf galaxies was 0% (all 8 were not assigned a pitch angle). Measurement success did not depend strongly on the M_b of the galaxies examined as measurement failures occurred at all magnitudes. However, no galaxies fainter than $M_b = -17$ magnitude were successfully measured. The characteristics of measured galaxies is included in table 1.

2.4. *Properties and Measurement of the Faint Sample*

Any comparison between two related populations of data requires an examination of which data, if any, is likely to be missing from each population. When the Davis et al. (2014) sample was assembled, it was chosen for reasons of sample completeness, namely, galaxies within its luminosity distance and M_b bounds were selected (bright enough to be observed and identified as spirals). We relax these conditions for the faint sample but partially recover completeness by statistically accounting for missing galaxies.

There presumably exist many galaxies within the distance and luminosity limits of the faint sample which are not observed due to the brightness limits of the CGS survey. This subsample of galaxies which are too faint to be detected in the CGS are missing from our population and need to be accounted for. In addition, we fail to measure pitch angles for some fraction of the sample, with spirals classified as m-type going completely unmeasured. The faint sample is morphologically different from Davis et al. (2014), containing an excess of c, cd, d, and m type galaxies and a dearth of a, ab, b, and bc types. Hubble types a and ab are nearly absent from the faint sample.

We separate the CGS by Hubble type into three groups to denote this populational difference in M_b, luminosity distance space (see Figure 3). We find a and ab galaxies to appear at $M_b = -19$ or brighter, with an average $M_b = -20.2$. Meanwhile, the cd, d, and m subgroup populates a lower brightness regime, with an average $M_b = -18.5$. Intermediate Hubble types span a large range of luminosities but have an average M_b consistent with the a and ab group. Even with m types removed, the cd and d group show a fainter average M_b.

Looking at the pitch angle measurement success rates in the faint sample, we find that m types have a success rate of zero, 67% for cd and d galaxies, and 85% for a, ab, b, bc, and c galaxies. Using Expectation-Maximization clustering, we divide the sample into galaxies with low pitch angle and galaxies with high pitch angle (see Figure 13). As expected, galaxies in the high pitch angle cluster are primarily cd and d types with a few c's. This low pitch angle cluster represents galaxies which went partially unreported in Davis et al. (2014).

Table 1. Measured pitch angles and corresponding black hole mass estimates for the CGS Faint sample galaxies.

Galaxy	Hubble Type	B_T	D_L	A_B	M_B	$\log(L/L_\odot)$	P	$\log(M/M_\odot)$
		[mag]	[Mpc]	[mag]	[mag]		[deg.]	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
ESO 219-G021	SABc	12.83	14.22	0.707	−18.64	9.59	-19.93 ± 2.36	7.69 ± 0.25
ESO 265-G007	SBc	12.54	13.4	0.498	−18.59	9.57	16.41 ± 2.12	7.69 ± 0.25
ESO 271-G010	Sc	12.90	18.889	0.355	−18.84	9.67	29.54 ± 9.45	7.68 ± 0.39
ESO 383-G087	Sd	11.70	2.675	0.257	−15.69	8.42	9.59 ± 0.76	7.77 ± 0.23
ESO 384-G002	SBd	12.88	17.45	0.232	−18.56	9.56	-11.89 ± 1.45	7.69 ± 0.25

Table 1 continued

Table 1 *(continued)*

Galaxy	Hubble Type	B_T	D_L	A_B	M_B	$\log(L/\mathrm{L}_\odot)$	P	$\log(M/\mathrm{M}_\odot)$
		[mag]	[Mpc]	[mag]	[mag]		[deg.]	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
ESO 479-G004	SBd	12.79	19.4	0.066	−18.72	9.62	28.12 ± 3.40	7.69 ± 0.25
IC 1954	Sb	12.10	15.287	0.059	−18.88	9.69	18.32 ± 2.70	7.68 ± 0.26
IC 1993	SABb	12.52	14.7	0.036	−18.35	9.48	-20.46 ± 2.48	7.70 ± 0.25
IC 2000	SBc	12.82	20.08	0.036	−18.73	9.63	-24.14 ± 1.93	7.69 ± 0.23
IC 2056	Sbc	12.80	20.5	0.067	−18.83	9.66	16.34 ± 5.06	7.68 ± 0.38
IC 2627	SABb	12.68	18.48	0.436	−19.09	9.77	-33.18 ± 23.55	7.68 ± 0.74
IC 5201	Sc	11.95	14.4	0.043	−18.88	9.69	34.92 ± 23.11	7.68 ± 0.70
IC 5273	SBc	12.74	16.557	0.045	−18.40	9.49	21.37 ± 6.88	7.70 ± 0.39
IC 5332	SABc	11.23	8.4	0.060	−18.45	9.52	4.99 ± 0.39	7.69 ± 0.23
NGC 24	Sc	12.14	6.889	0.071	−17.12	8.99	-25.40 ± 23.11	7.73 ± 0.94
NGC 45	SABd	11.39	9.313	0.075	−18.53	9.55	-27.98 ± 3.97	7.69 ± 0.26
NGC 247	SABc	9.72	3.587	0.066	−18.12	9.38	42.90 ± 4.84	7.70 ± 0.24
NGC 300	Scd	8.81	1.972	0.046	−17.71	9.22	33.69 ± 0.75	7.72 ± 0.22
NGC 701	SBc	12.84	22.711	0.091	−19.03	9.75	17.15 ± 1.78	7.68 ± 0.24
NGC 779	SABb	12.27	17.678	0.097	−19.06	9.76	15.58 ± 5.52	7.68 ± 0.41
NGC 1022	SBa	12.32	18.5	0.093	−19.11	9.78	22.44 ± 12.10	7.68 ± 0.58
NGC 1042	SABc	12.11	9.431	0.104	−17.87	9.28	20.71 ± 6.55	7.71 ± 0.38
NGC 1179	Sc	12.83	18.171	0.087	−18.55	9.56	-19.18 ± 5.32	7.69 ± 0.35
NGC 1249	SBc	12.78	15.854	0.060	−18.28	9.45	-17.13 ± 5.93	7.70 ± 0.41
NGC 1292	Sc	12.79	21.475	0.062	−18.93	9.71	-14.87 ± 3.37	7.68 ± 0.31
NGC 1313	SBcd	9.66	3.951	0.396	−18.72	9.62	-41.63 ± 1.08	7.69 ± 0.22
NGC 1337	SBc	12.53	16.075	0.245	−18.75	9.63	-17.57 ± 5.26	7.69 ± 0.37
NGC 1436	Sb	12.86	18.36	0.039	−18.50	9.53	-24.58 ± 4.48	7.69 ± 0.28
NGC 1487	Scd	12.33	8.9	0.043	−17.46	9.12	-69.44 ± 2.21	7.72 ± 0.22
NGC 1493	SBc	11.72	11.3	0.037	−18.58	9.57	14.21 ± 3.67	7.69 ± 0.34
NGC 1494	Scd	12.10	15.217	0.022	−18.83	9.67	22.01 ± 2.05	7.68 ± 0.24
NGC 1518	SBd	12.26	9.596	0.174	−17.82	9.27	-46.01 ± 9.83	7.71 ± 0.30
NGC 1637	Sc	11.63	10.703	0.146	−18.66	9.60	13.76 ± 1.68	7.69 ± 0.25
NGC 1688	SBcd	12.20	15.45	0.125	−18.87	9.68	-38.66 ± 0.95	7.68 ± 0.22
NGC 1744	SBc	11.71	10.853	0.148	−18.62	9.58	10.61 ± 3.75	7.69 ± 0.41
NGC 1796	SBc	12.88	10.6	0.088	−17.33	9.07	15.02 ± 2.92	7.73 ± 0.29
NGC 2082	SBb	12.73	17.978	0.210	−18.75	9.64	22.26 ± 2.41	7.69 ± 0.24
NGC 2090	Sbc	11.66	12.758	0.144	−19.01	9.74	8.29 ± 2.37	7.68 ± 0.36
NGC 2427	SABd	12.30	11.782	0.778	−18.83	9.67	20.87 ± 3.62	7.68 ± 0.28
NGC 3513	SBc	12.04	13.153	0.23	−18.79	9.65	28.76 ± 0.56	7.68 ± 0.22
NGC 4592	Sd	12.9	11.63	0.082	−17.51	9.14	40.30 ± 17.93	7.72 ± 0.49
NGC 4632	Sc	12.63	18.518	0.087	−18.79	9.65	22.92 ± 3.74	7.68 ± 0.27
NGC 4951	SABc	12.58	16.6	0.171	−18.69	9.61	18.51 ± 5.94	7.69 ± 0.39
NGC 5068	Sc	10.64	6.075	0.369	−18.65	9.59	23.60 ± 13.00	7.69 ± 0.59

Table 1 continued

Table 1 *(continued)*

Galaxy	Hubble Type	B_T	D_L	A_B	M_B	$\log(L/\mathrm{L}_\odot)$	P	$\log(M/\mathrm{M}_\odot)$
		[mag]	[Mpc]	[mag]	[mag]		[deg.]	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NGC 5134	SABb	12.46	10.889	0.328	−18.05	9.36	-36.28 ± 5.67	7.71 ± 0.27
NGC 5556	Scd	12.81	18.75	0.254	−18.81	9.66	-14.09 ± 1.31	7.68 ± 0.24
NGC 7361	Sc	12.82	17.2	0.06	−18.42	9.50	24.12 ± 7.22	7.70 ± 0.37
NGC 7412	SBb	11.92	12.485	0.042	−18.60	9.58	14.66 ± 2.23	7.69 ± 0.26
NGC 7421	Sbc	12.78	22.6	0.054	−19.04	9.75	20.88 ± 7.18	7.68 ± 0.41
NGC 7424	Sc	11.54	11.5	0.039	−18.80	9.66	20.24 ± 6.05	7.68 ± 0.37
NGC 7456	Sc	12.43	16.185	0.039	−18.65	9.60	-18.48 ± 7.27	7.69 ± 0.45
NGC 7496	Sb	12.9	15.02	0.036	−18.02	9.34	-36.03 ± 0.84	7.71 ± 0.22
NGC 7713	Scd	11.87	10.272	0.06	−18.25	9.43	-21.62 ± 2.78	7.70 ± 0.25
NGC 7793	Scd	9.77	4.171	0.071	−18.40	9.50	11.37 ± 1.21	7.70 ± 0.24
PGC 3853	SABc	12.62	12.6	0.48	−18.36	9.48	23.17 ± 5.45	7.70 ± 0.32

NOTE—Columns: (1) Galaxy Name. (2) Hubble type morphological classification. (3) B-Band total magnitude. (4) Luminosity distance (in Mpc). (5) B-Band extinction magnitude from Schlafly & Finkbeiner (2011). (6) B-Band absolute magnitude. (7) Luminosity. (8) Pitch angle (in degrees). (9) Black hole mass.

2.5. *Pearson Distribution Fitting of Pitch Angles*

In Davis et al. (2014), the probability density of pitch angles for their sample was calculated by creating a "binless" histogram which treats the dataset as a sum of Gaussians, with individual P measurements and their error bars representing the means and standard deviations, respectively. This histogram was then statistically fit to a probability density function (PDF) based on the statistical properties of the dataset. This skew-kurtotic-normal fit of the Pitch Angle Distribution could then be transformed into a BHMF through the M–P relation (Seigar et al. 2008; Berrier et al. 2013; Davis et al. 2017). We reproduce these calculations for the current dataset (Figure 7).

The skew-kurtotic-normal fit of the sample does not match well with the binless histogram of the sample. Low N in the faint sample may be a partial cause of irregularities between the histogram and the PDF. Furthermore, this disparity may be due to difficulties in measuring these fainter or lower contrast galaxies, as their higher associated errors potentially alter the shape of the histogram. Manually creating a best fit PDF to match the binless histogram required much higher values of both skewness and kurtosis than the original fit from the statistics of the sample. Given the peak locations of the bright sample and faint galaxies sample match up reasonably well, the increased skew in the faint galaxies sample is interpreted mainly as a stronger tail of probability density at high (past peak) pitch angles. Moreover, excess kurtosis in the faint galaxies sample is indicative of a heavy tail to the distribution.

The excess density at the peak may be related to the excess of $m = 2$ spirals in the faint sample. Galaxies with two-arm symmetry tend toward successful pitch angle measurement as compared to flocculent galaxies or galaxies with higher-order arm symmetry. The two-symmetry galaxies are often more grand design-like, with higher arm-interarm contrast whereas galaxies assigned $m = 4$ or $m = 6$ symmetry have a tendency toward clumpiness, with an agglomeration of many arm segments and spurs rather than sharply defined, contiguous arms. Difficulties associated with the measurement of flocculent galaxies in the sample may bias the measurement toward the peak of the distribution.

Another possible cause of the excess at peak for the faint sample is the measurement success rate by Hubble type, paired with the Hubble type abundance in the sample. As compared to the Davis et al. (2014) sample, ours has fewer a-bc Hubble types and an excess of c-m types. The c-m types more commonly evade pitch angle measurement. Also, the peak excess might be the result of the relative lack of a and ab Hubble types in the faint sample. The absence of these higher mass spiral galaxies from the distribution would naturally raise the probability densities estimated for all other masses. The combination of a and ab types largely missing from the sample and cd and d types more commonly evading pitch angle measurement might account for the lower probability

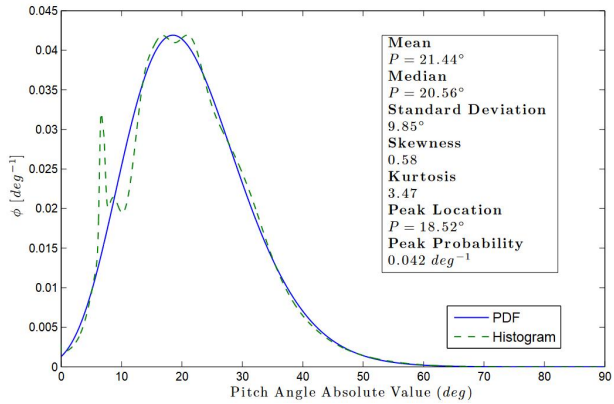
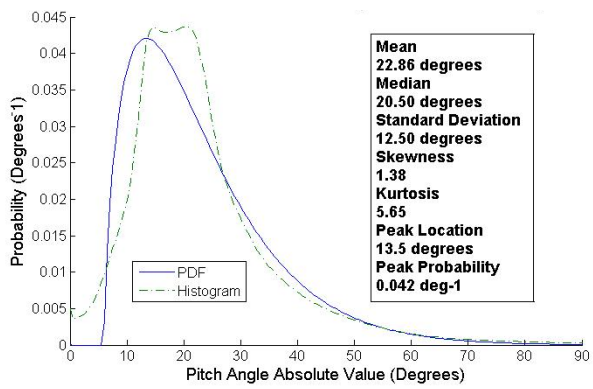
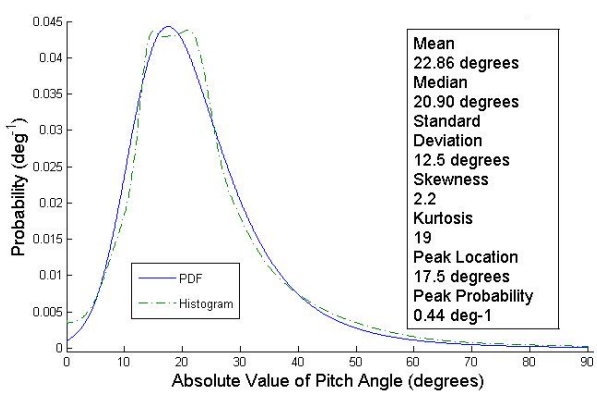


Figure 7. Above: Pitch angle distribution function from Davis et al. (2014). Below: PADFs for the faint galaxies sample. Solid lines are the Probability Density Function fits. Dotted lines are the distribution of P, modeled as a "binless" histogram, where each data point is a Gaussian with mean equal to pitch angle measured and standard deviation equal to associated measurement error. The traditional Probability Density Function fit does not match well to the data. A second attempt at best fitting a skew-kurtotic-normal distribution to the data is shown at the bottom. The faint galaxies sample shows higher skewness and kurtosis than its higher luminosity counterpart, indicating data points with larger associated errors and a heavier tail to the distribution. Graphs generated using Becker (2010)



.



densities on either side of the peak of the pitch angle distribution function (PADF) and a corresponding relative amplification of the peak. When combined with the other previously mentioned causes, it is not all that surprising that the fainter galaxy sample would show a higher peak probability than its brighter counterpart. Finally, The Fundamental Plane, as reported in Davis et al. (2015), would support the idea of smaller bulges of these later type galaxies being compatible with tighter pitch angles if they had low gas densities in their disks. This is a prospective subject for further study.

Cullen and Frey graphs are used to compare functions in skewness-kurtosis space. They are produced for each sample (Delignette-Muller & Dutang 2015) to further scrutinize distribution shape in the context of these parametric solutions (see Figure 8). Possible values of the skewness and kurtosis are calculated by bootstrapping (here, boot=2000) to illustrate the likely functional form of the parent distribution. The outcome is the bright sample more closely resembling a normal distribution than the faint galaxies sample. The excess skewness and kurtosis of the faint sample as interpreted in this way are somewhat reduced. The statistical fit of these values to the data did not match well to the binless histogram, and scatter in bootstrapped values for the faint sample is much larger. Accordingly, differences retained between the samples at the tail of the distribution are not ruled out. Consequently, we perform non-parametric fitting of the samples to further compare their tail behavior.

3. PITCH ANGLE DISTRIBUTION

In order to assess whether the BHMF for the faint sample's lower luminosity regime of galaxies resembles those in preceding studies of brighter galaxies, an effective tool is to examine the Pitch Angle Distribution Function (PADF) of these galaxies in comparison to prior work. If the current set of galaxies comprise the same population as their brighter counterparts, their PADFs will mirror one another, whereas if the current sample encompasses a second population of galaxies (presumably at the low mass end of the distribution) this would alter the shape of the BHMF in this regime.

3.1. *Non-Parametric Fitting*

While parametric solutions to distribution fitting are simpler to express in functional forms, with insufficient data they are sometimes overly constrained. It is clear the initial Pearson fit to the faint sample was poor, likely due to the low number of galaxies in the sample. In such a low N case, a non-parametric Kernel Density Estimator (KDE) approach is more appropriate for PDF

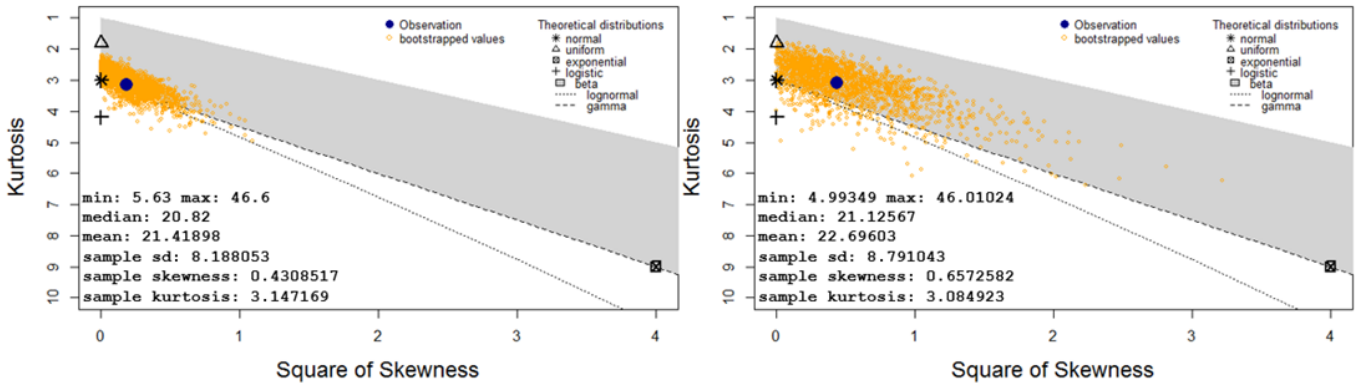


Figure 8. Comparison of Cullen and Frey graphs with a 2000 sample bootstrap for the Pearson Distributions of the Davis et al. (2014) sample (left) and faint galaxies sample (right). Despite the Pearson Distribution for the faint sample not fitting well to its data (i.e., not enough skewness or kurtosis), it is centered farther from the normal distribution on the Cullen and Frey graphs, illustrating this difference between the two samples.

production and dataset comparison (see Figure 9). The distribution of the Davis et al. (2014) sample is closer to Gaussian, due to both morphology trends and to the larger sample size of the Davis distribution. For fuller assessment, KDE is chosen to fit both datasets.

Even though the two samples are morphologically distinct, their pitch angle distributions are appreciably similar. A possible tendency towards very loose spirals at our distribution's tail betrays a sign of the shift to higher P one may presume in a sample dominated by c and d Hubble types. This bump in the tail of the distribution is small, and measurement errors for 2DFFT are known to be larger in higher P regimes. As such, further analysis (below) is required to ascertain the strength of the tail.

Kernal Density Estimator (KDE) fits of the two distributions are performed as a method of comparing the samples. As with the parametric (Pearson Distribution) fitting solutions, both samples have very similar peak pitch angles. The bright sample shows a broader and lower peak with excess probability slightly past peak but deficit probability far past peak in the high pitch angle regime. This naive gaussian KDE, however, does not fully take into account some of the behaviors of the data which might affect the probability distribution. Among these factors to correct for are bandwidth selection in performing the KDE and weighting for heteroscedasticity in the data. Confidence intervals around the best fit KDE are also requisite to attain a comparison of samples which best shows their compatibility.

3.2. *Pitch Angle Dependent Errors and Bandwidth Optimization*

3.2.1. *Heteroscedasticity*

Heteroscedasticity is an occurrence where the variance of a dataset changes for various subpopulations of the dataset. A common case of this is where the magnitude of the errors in one variable are covariant with the magnitude of another of the variables being examined. In order to properly KDE fit the PDFs of the two samples, heteroscedasticity of the datasets is assessed.

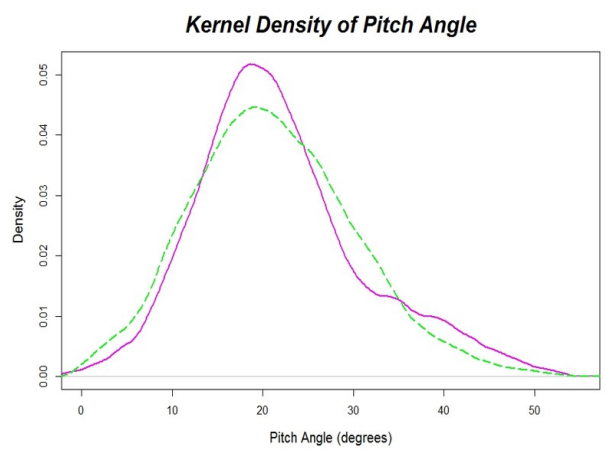


Figure 9. Comparison of Initial Kernel Density Estimator fits of the Probability Densities for the bright sample (green dashed) and faint galaxies sample (magenta solid). The faint galaxies sample has a slightly higher peak and heavier tail than its counterpart. Both distributions are strikingly similar at the lower pitch angle (higher mass) end of the function. The two functions reach peak probability density at nearly the same pitch angle. Graphs generated with Guidoum (2015)

.

In the case of galactic pitch angle measurement, recorded errors are expected to grow with pitch angle. Spiral galaxies with small pitch angles (tightly wound) tend to be measured with more accuracy than those with large pitch angles (loosely wound). One reason for this error growth with pitch angle is largely the result

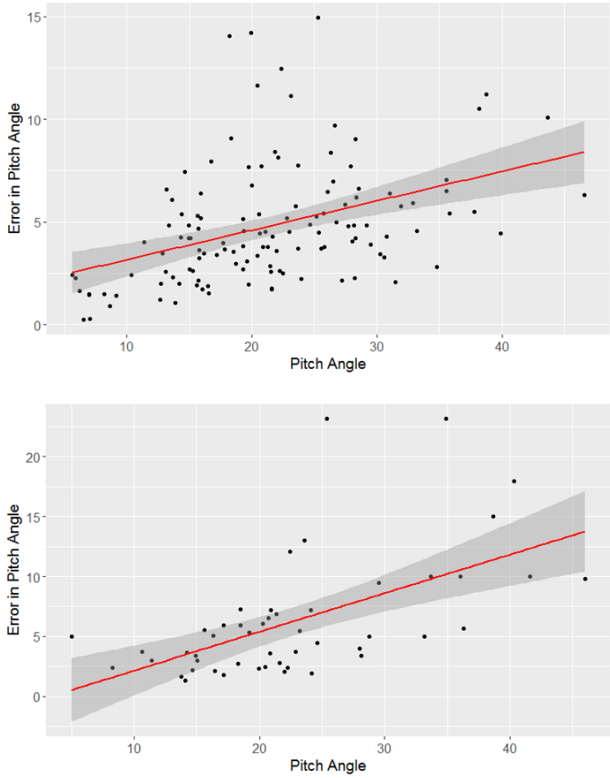


Figure 10. Measurements of Pitch Angle are heteroscedastic, for higher absolute values of pitch angle, the associated error in measurements is larger. Above: Error in P as a function of P for the Davis et al. (2014) sample. Below: Error in P as a function of P for the faint galaxies sample. Linear fits of heteroscedasticity are the red lines, 95% confidence intervals are shaded. Both data sets show increasing measurement errors as a function of pitch angle. Graphs generated using Fox & Weisberg (2011).

of the measurement technique employed. Galaxies with tight spirals have a broader swath of phase angles (more rotations) of spiral arm to measure, while very loose spirals might not have even one measurable rotation.

It is shown in Figure 10 that both samples do have errors which grow as a function of pitch angle, in accordance with expectations. A linear model is fit to these errors. Robust Regression is applied in order to assign weights which minimize the impact of outliers to the fit. From this linear fit to the heteroscedasticity of the data, Huber weights are calculated which may be applied to each galaxy in distribution fitting. In this way, the excess error on the tail of the distribution is accounted for in the determination of the probability density.

3.2.2. *Bandwidth Selection*

Bandwidth selection is required to produce a more correct KDE. Maximum Likelihood Cross Validation (MLCV) assesses the optimal bandwidth for the data. Bandwidth in KDE may be thought of as a smoothing parameter of the fit when the number of observations is low. Higher bandwidths create functions with more smoothing while smaller bandwidths correspond to fits which are allowed more local variance. To avoid over or under smoothing the fit, we explore the effects of bandwidth selection on the behavior of the distribution. While peak height of the faint sample varies with bandwidth, the tail behavior of the distribution is largely bandwidth independent (see Figure 11). This means we can trust the tail behavior of interest to be similar regardless of the bandwidth chosen, though the MLCV bandwidth is likely to be the most representative of the sample.

Using our MLCV-generated bandwidth with our Robust Weighted, Heteroscedasticity-corrected samples, we convert from Pitch Angle to Black Hole Mass using the M–P relation and produce a KDE of the BHMF for the two samples (see Figure 12). Errors are assessed using bootstrapping to find the 90% confidence interval around each density estimate. Another alternate technique tested is Adaptive Bandwidth Kernel Density Estimation, wherein the shape of the kernel used to produce the density is allowed to change shape (broaden or tighten) based on local properties of the data. Adaptive bandwidth techniques are advantageous when estimating density for heavy tailed or multimodal distributions such as our faint galaxies sample. Confidence intervals are assessed around each sample using a 10000 sample bootstrap of the density estimation.

4. BLACK HOLE MASS FUNCTION

The PADF is transformed to a BHMF using the M–P relation as described in Berrier et al. (2013). Error prop-

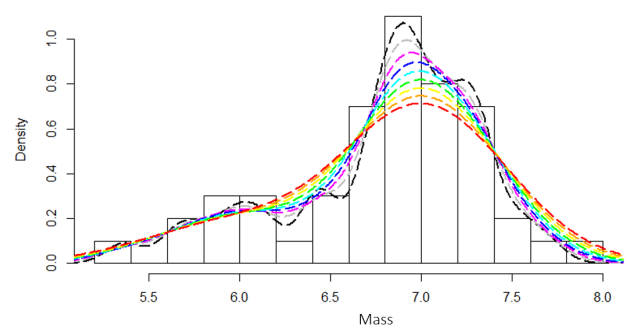


Figure 11. Bandwidth Comparison for Faint Sample. At higher bandwidths, the Faint Sample closer resembles the previous sample, while at lower bandwidths, some differences emerge. Bandwidth optimization techniques yield a bandwidth of 0.22 as optimal. Bandwidth selection does not strongly impact the height of the low mass tail of the distribution, and most strongly affects the density peak. Bandwidths: (0.1, 0.13, 0.16, 0.19, 0.22, 0.25, 0.28, 0.31, 0.34).

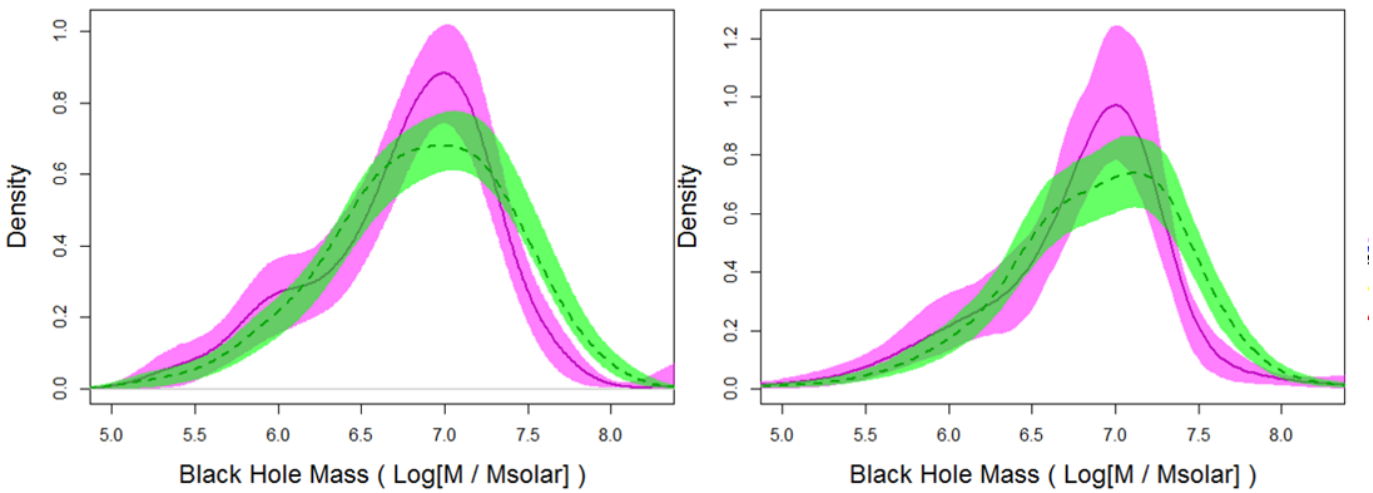


Figure 12. Left: Faint sample (magenta, solid line)and Davis et al. (2014) sample (green, dotted line) BHMF's as fit with Bandwidth Selected Kernel Density Estimator (bw=0.22) and robust weighting. Right: Adaptive Bandwidth Kernel Density Estimator BHMF's. 90% confidence interval found using a 10000 sample bootstrap is shown for each estimate. The faint sample shows a slight excess at the low mass tail of the distribution while the Davis et al. sample shows an excess at the high mass end which is likely due to more a and ab type galaxies in their sample.

agation is performed before distribution fitting for the BHMF of each sample. Careful treatment of KDE produces updated estimates of the BHMF for these sets of spiral galaxies in the local universe. The distribution of masses for the faint galaxies sample largely mirrors that of Davis et al. (2014), with the anomalies mentioned earlier in the description of the PADF translated to mass space, namely, excess of density at the peak and somewhat for the tail for the faint sample, and under density of the faint sample at black hole masses higher than the mass of peak probability.

4.1. *Pitch Angle Clustering*

In order to further understand the behavior of the faint sample, various clustering techniques are employed. As the tail of the distribution of faint sample galaxies is heavy, and there is some indication of population differences between the Faint Sample and its brighter counterpart despite similar peak behavior, it is useful to explore whether the tail of the faint sample represents a second population. We perform E-M clustering analysis of the pitch angle distribution to divide the faint sample into a High Pitch and Low Pitch subsample. Simple normality tests are not conclusive as to whether there is a true second population within the sample. However, examining the coincidence of population sorting with Hubble type reveals the high pitch galaxies to consist of c, cd, and d hubble types. Not all of the c, cd, and d types were sorted into the high pitch cluster but none of the a-bc types were assigned to the high pitch cluster. From this examination it is shown some fraction of the very late Hubble types compose the second population. If the distribution is to be treated as one function rather than two, the single population carries a heavy tail made up of these latest hubble types.

To further flesh out this relationship, we fit the KDE probability densities of the sample by hubble type. We find the cd and d type galaxies to indeed peak at a higher pitch angle (lower SMBH mass). If the measured galaxies in the faint sample are divided by hubble type, c types make up half the sample at 25 measured galaxies, with a, ab, b, bc together making up about a quarter (all but one measured being b, bc) and cd,d,m making up the final quarter (no m galaxies were measured). When a, ab, b, bc, c are grouped together and cd,d,m are grouped together we find the tighter pitch angle, high mass group is well behaved with a very weak tail, but the cd,d group has a broad peak shifted to higher pitch angles (max P=23.24 degrees, mean P=28.69 degrees). This is not unexpected, as hubble types are defined in part by the tightness of spiral arm winding. Hubble type letters past c should indeed correspond to loose spirals.

Using the two clustering populations, we produce a possible two-population model for the probability density which matches closely with the global population. The faint Sample population may be described with the superposition of two near-normal distributions or as one population with a heavy tail. In either case, the behavior at the extreme high P / low M end of the distribution is explained by the cd and d type galaxies in the sample.

4.2. *Combined BHMF for Local Spiral Galaxies Accounting for Sample Completeness*

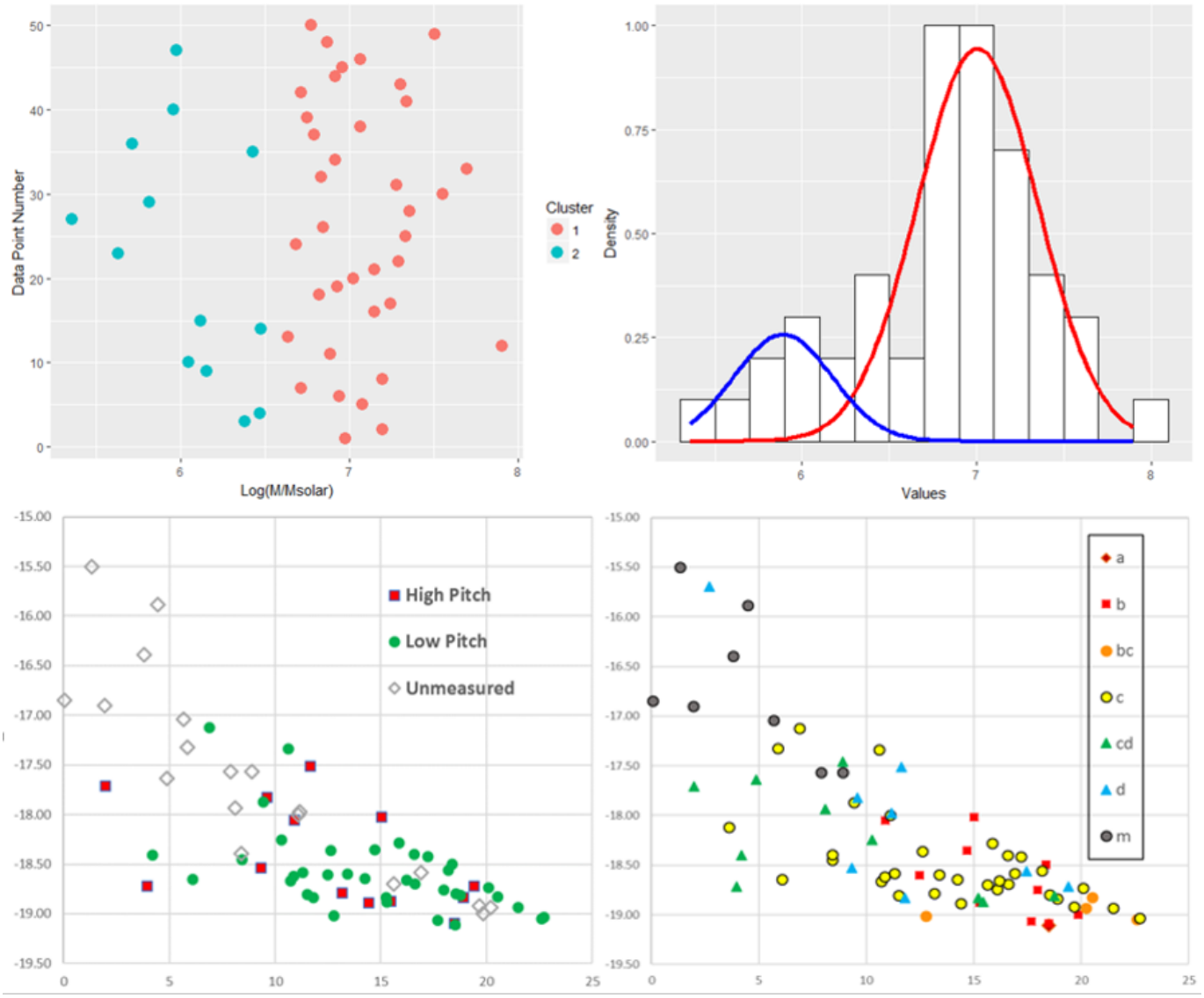


Figure 13. Top-left: Clustering of the faint sample into two clusters in black hole mass. Top-right: Histogram overlaid with distribution fits for each cluster. Graphs generated using Wickham (2009). Bottom-left: Faint sample galaxies plotted by cluster. Low pitch / high mass (green circles), high pitch / low mass (red squares), and unmeasured pitch angles (gray diamonds). Bottom-right: Faint sample by Hubble type. Note: m dwarf Hubble types comprise nearly all the galaxies fainter than -17 abs. B mag. All of these galaxies went unmeasured in pitch angle.

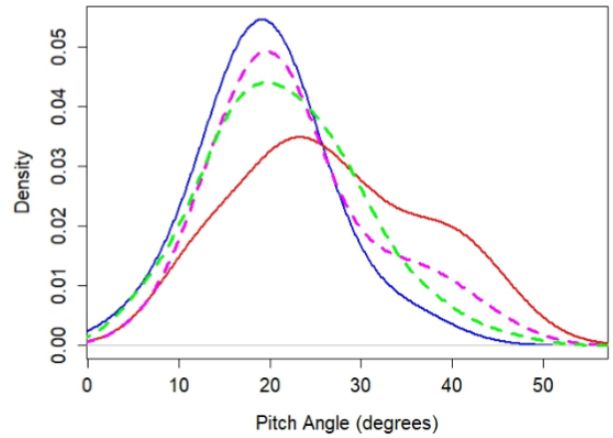
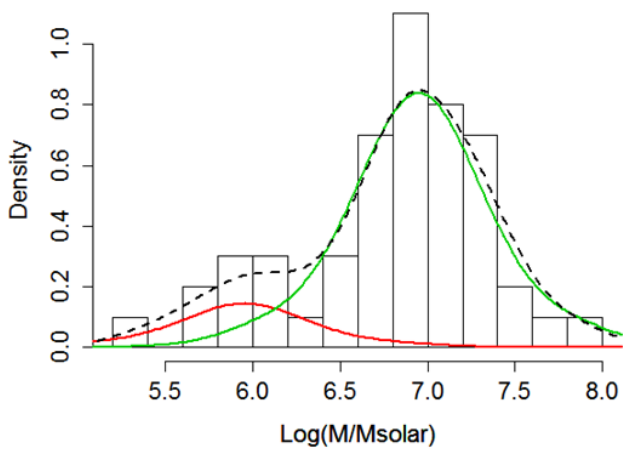


Figure 14. Above: Corrected PADFs of key subsamples. Green: bright limited Sample. Magenta: Faint Sample. Blue: Faint subsample of Hubble types a-c. Red: Faint subsample of Hubble types cd,d. From this figure it is clear that while the cd,d type galaxies make up a small fraction of the faint sample, they are the cause of most of the behavior at the tail of the distribution. It is also apparent that the excess peak height in the faint sample is a result of the comparative overabundance of Hubble Type c galaxies, which cluster strongly around the peak. Below: BHMF of the faint sample decomposed into a low-mass and high mass-population. Curves are semiparametric fits to the cluster data. Dotted line is KDE of original distribution.



In order to construct the BHMF of local spiral galaxies we combine the bright sample from earlier work with the faint sample to eliminate the M_b constraint on the mass function. To do this properly, galaxies missing from the faint sample due to detection limits must be accounted for. This is accomplished by reweighting the galaxies from both samples based on the effective comoving volume of galaxies by M_b. The fainter the galaxies targeted, the smaller effective comoving volume in which such a galaxy could be detected. Our corrected Luminosity Function uses a Pearson Distribution fit of the number of galaxies by magnitude with the effective comoving volume to up-correct the number of galaxies at very faint magnitudes. We check the Pearson Distribution Fit against a KDE and find a reasonable match to the data, providing confidence in the computed Luminosity Function.

With this new weighting based on effective comoving volume it is possible to properly combine the complete sample and adjusted incomplete sample into one population which describes the Black Hole Mass Function in the local universe. The BHMF is calculated in the same manner as with each individual sample, but with a reweighting of the combined input data to attempt to account for missing galaxies. The result of this combination retains some of the properties of each of the samples from which it originated, as it is effectively a weighted average of the two distributions. As such, the peak location remains largely unchanged. There is some retention of a heavier tail to the distribution at high P / low SMBH mass. This BHMF describes the distributions of non-dwarf spiral galaxies in the local universe.

5. DISCUSSION

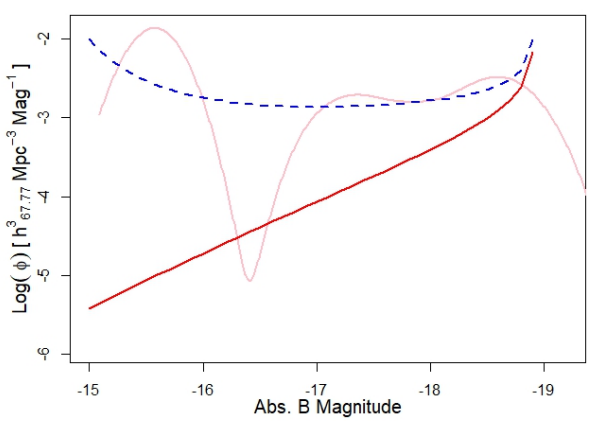


Figure 15. Luminosity Functions for CGS Faint Galaxies. Red: Uncorrected Luminosity Function of the Sample. Blue: Luminosity Function Fit as corrected for effective comoving volume of the galaxies in the sample. Pink: KDE check of the corrected Luminosity Function, showing the probability density based on actual galaxies. The number of measured galaxies at fainter than $M_b = 17$ was very small, accounting for the spikes below and above the functional fit correction. The corrected Luminosity Function produces reasonable galaxy abundances for the corrected sample. Galaxies in this regime are reweighted for density calculations with this correction to produce the BHMF for all measurable local spiral galaxies.

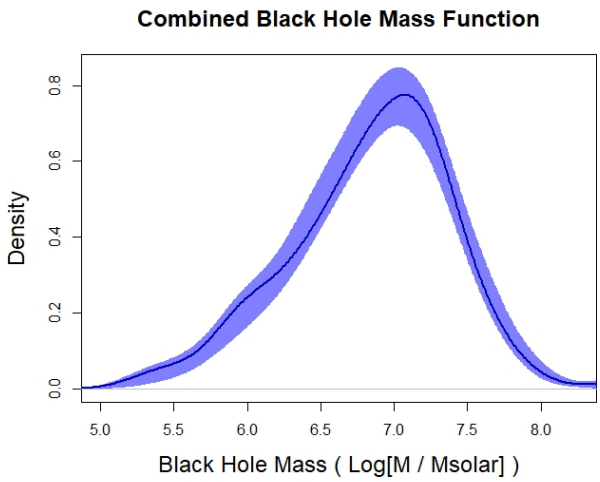


Figure 16. Combined Black Hole Mass Function with 90% confidence intervals for CGS galaxies with limiting luminosity distance of 25.4 Mpc. This Mass Function is representative of all spiral galaxies of Hubble Types a-d (m type spiral dwarf galaxies were not measured with any accuracy).

The PADF and BHMF established from our latest sample of local disk galaxies resembles Davis et al. (2014). These subsets are mirrored at low pitch angles / high masses. However, they may diverge at the high pitch angle end of the dataset. The faint sample has a higher probability of large pitch angles (loose spirals) and a reduced probability of containing small pitch angles (tight spirals). The difference between the two samples as denoted by our KDE approach is small, and the errors and biases in the pitch angle measurement process are not yet well enough defined to make an assertive distinction (see Figure 14). However, the morphological differences between the galaxies on the heavy tail of the distribution and those at the peak do suggest two separate populations.

The CGS sample contains a difference in Hubble type across abs. B mag. space, which illustrates the major difference between the two samples in question. The faint sample is largely missing its a and ab types and as a result has an excess of all other types. The Davis et al. (2014) sample, meanwhile, is missing the m types and perhaps some cd and d types as well, given the luminosity limit. The excesses and scarcities of the samples are borne out in density estimation of their PADFs and BHMFs. In comparison to the Davis et al. (2014) sample, the faint sample expresses scarcity at low P (high M) and overabundance at the distribution peak, with hints of slight overabundance at the distribution tail. The tail of the faint sample is heavy, suggesting the possibility of a multimodal distribution. We show that the cd and d galaxies cluster apart from the a and ab types in M_b, and largely comprise the 'high pitch angle' subgroup found by two group clustering. Removal of this subgroup and replacement with typical a and ab galaxies from the faint sample would likely cause the faint sample and the Davis et al. (2014) sample to have convergent BHMFs.

As a whole, the resulting BHMFs of our sample and Davis et al. (2014) may be construed to be nearly identical once differences in population subgroups are accounted for. The weight of the low-mass tail of the BHMF remains somewhat unknown. Due to missing faint galaxies and poor pitch angle measurement rates in this regime, we are likely to be underestimating the density of the BHMF at its tail. The pitch angle measurement success rate for a-c types was better than the cd,d,m tpye galaxies. Since the distribution for cd,d types peaks at higher pitch angles, it follows the tail of the distribution may be heavier than we measure. The extreme low-mass end (m types and other dwarfs) remains unmeasured to date. By correcting our sample for effective comoving volume, we attempt to recover a more representative distribution of galaxy pitch angles by accounting for galaxies missed due to their low surface brightness. In so doing, a combined sample is produced containing all galaxies within the CGS whose pitch angles were measured and whose limiting luminosity-distance was 25.4 Mpc. This combined sample is representative of the Local BHMF of spiral galaxies (excluding Magellanic spirals) in the local universe.

The lowest mass end of the spiral galaxy portion of the population is a strong candidate set for further study toward characterizing Intermediate Mass Black Holes. Graham et al. (2019) examine a sample of 74 IMBH candidates in Virgo Cluster galaxies with three different black hole mass scaling relations and find 33 galaxies predicted to have masses less than $10^5 - 10^6 M_\odot$ range, with many in the $10^4 - 10^5 M_\odot$ range. Late type spirals with very loose (high pitch angle) arms represent a good candidate set for further IMBH study. Spiral galaxies with Pitch Angles greater than 27 degrees correspond to masses $< 10^5 M_\odot$ and likely fall into the IMBH range (Graham et al. 2019). However, errors in pitch angle measurement at this pitch angle regime tend toward 5-10 degrees in pitch, which corresponds to more than an order of magnitude in mass. Precise mass estimates for these galaxies are challenging without the application in tandem of other black hole mass scaling relations for late type galaxies. Dwarf galaxies are also promising candidates for IMBHs, although those dwarfs without clear spiral structure must rely solely on other mass scaling relations for measurement. Filling the gaps in knowl-

edge surrounding this low mass regime continues to be a desirable result.

6. CONCLUSIONS

The latest sample of CGS spiral galaxies have a pitch angle distribution function (and corresponding SMBH BHMF) analogous to that of Davis et al. (2014). There is strong indication of high-mass black holes missing from the faint sample, and some indication of an excess of lower mass black holes on the distribution tail. It may be claimed although populational differences exist between the two samples, the BHMF of spiral galaxies in the local Universe is not strongly altered at its low-mass end by the inclusion of galaxies with cd and d Hubble types. Excesses at the tail of the BHMF may be larger than recorded in our data, due to galaxies too faint to be observed and unmeasurable pitch angles for latest Hubble types. Davis et al. (2017) claim that galaxies with pitch angles greater than 26.7° are candidates for hosting Intermediate Mass Black Holes (masses $< 10^5 M_\odot$). It may be the case that the heavy tail of our pitch angle distribution is composed in part of galaxies hosting these objects.

Sellwood & Carlberg (2019) provide evidence of self-excited instabilities as the cause of spiral patterns in simulations of unperturbed disks. They find that although individual spiral modes are transient, fresh instabilities are excited as the old modes fade away, resulting in a recurrent instability cycle. If spirals are primarily driven by density waves, the newly birthed modes should have the same pitch angle as their dying parents. This has been shown to be the case in some simulations (**?**). If this is the case, pitch angle measurement as a method for probing other galaxy characteristics is likely to be robust for a given galaxy across time, perhaps with the exception of transitional periods between dominant modes.

There exists a planar relationship among the quantities of central bulge mass, pitch angle, and galactic neutral hydrogen gas density (Davis et al. 2017). It has also been shown that higher disk masses favor lower multiplicity spiral patterns (Berrier & Sellwood 2015; Sellwood & Carlberg 1984; Athanassoula et al. 1987). Galaxies with lower disk densities should, then, have a tendency toward higher number of arms and less m=2 grand design spirals. These lower disk density galaxies could produce spirals with similar pitch angles to their heavy disk counterparts if their bulges are less massive. Going forward, A census of neutral hydrogen gas density in galaxies targeted for pitch angle measurement should be taken, such that the populational differences among these parameters may be better explored. This is especially relevant for galaxies without grand design spiral structure, as their disks are likely to be less massive. In order to accurately produce the BHMF for late type galaxies, this alteration of pitch angle due to different gas densities should be taken into account.

The performance of pitch angle measurement techniques as a function of galaxy morphology, luminosity, and redshift, is yet to be surveyed. Studies in these areas, as well as in selection effects and biases, should reduce measurement errors and improve measurement success rates for future work. For now, we largely reaffirm the BHMF for local spiral galaxies Davis et al. (2014), with the possibility of another population of lower mass black holes at the very late Hubble type, high pitch angle tail of the distribution.

REFERENCES

Athanassoula, E., Bosma, A., & Papaioannou, S. 1987, A&A, 179, 23

Bae, J., Nelson, R. P., & Hartmann, L. 2016, ApJ, 833, 126

Becker, M. 2010, PearsonDS: Pearson Distribution System, r package version 0.92. http://CRAN.R-project.org/package=PearsonDS

Berrier, J. C., & Sellwood, J. A. 2015, ApJ, 799, 213

Berrier, J. C., Davis, B. L., Kennefick, D., et al. 2013, ApJ, 769, 132

Cuzzi, J. N., Lissauer, J. J., & Shu, F. H. 1981, Nature, 292, 703

Dalla Bontà, E., Ferrarese, L., Corsini, E. M., et al. 2009, ApJ, 690, 537

Davis, B. L., Berrier, J. C., Shields, D. W., et al. 2012, ApJS, 199, 33

—. 2016, 2DFFT: Measuring Galactic Spiral Arm Pitch Angle, Astrophysics Source Code Library, , , ascl:1608.015

Davis, B. L., Graham, A. W., & Seigar, M. S. 2017, MNRAS, 471, 2187

Davis, B. L., Berrier, J. C., Johns, L., et al. 2014, ApJ, 789, 124

Davis, B. L., Kennefick, D., Kennefick, J., et al. 2015, ApJL, 802, L13

Delignette-Muller, M. L., & Dutang, C. 2015, Journal of Statistical Software, 64, 1. http://www.jstatsoft.org/v64/i04/

Elmegreen, B. G., Elmegreen, D. M., & Montenegro, L. 1992, ApJS, 79, 37

Ferrarese, L., & Merritt, D. 2000, ApJL, 539, L9

Fox, J., & Weisberg, S. 2011, An R Companion to Applied Regression, 2nd edn. (Thousand Oaks CA: Sage). http://socserv.socsci.mcmaster.ca/jfox/Books/Companion

Franceschini, A., Vercellone, S., & Fabian, A. C. 1998, MNRAS, 297, 817

Gebhardt, K., Bender, R., Bower, G., et al. 2000, ApJL, 539, L13

Graham, A. W., Driver, S. P., Allen, P. D., & Liske, J. 2007, MNRAS, 378, 198

Graham, A. W., Scott, N., & Schombert, J. M. 2015, Publication of Korean Astronomical Society, 30, 335

Graham, A. W., Soria, R., & Davis, B. L. 2019, MNRAS, 484, 814

Guidoum, A. 2015, kedd: Kernel estimator and bandwidth selection for density and its derivatives., r package version 1.0.3. http://CRAN.R-project.org/package=kedd

Ho, L. C., Li, Z.-Y., Barth, A. J., Seigar, M. S., & Peng, C. Y. 2011, The Astrophysical Journal Supplement Series, 197, 21

Lin, C. C., & Shu, F. H. 1964, ApJ, 140, 646

Marconi, A., Risaliti, G., Gilli, R., et al. 2004, MNRAS, 351, 169

Natarajan, P. 2011, ArXiv e-prints, arXiv:1105.4902

Peng, C. Y., Ho, L. C., Impey, C. D., & Rix, H.-W. 2010, AJ, 139, 2097

Pérez, L. M., Carpenter, J. M., Andrews, S. M., et al. 2016, Science, 353, 1519

Puerari, I., Block, D. L., Elmegreen, B. G., Frogel, J. A., & Eskridge, P. B. 2000, A&A, 359, 932

Richstone, D., Ajhar, E. A., Bender, R., et al. 1998, Nature, 395, A14

Richstone, D. O. 2002, Reviews in Modern Astronomy, 15, 57

Rodríguez, S., & Padilla, N. D. 2013, MNRAS, 434, 2153

Salucci, P., Szuszkiewicz, E., Monaco, P., & Danese, L. 1999, MNRAS, 307, 637

Savorgnan, G., Graham, A. W., Marconi, A., et al. 2013, MNRAS, 434, 387

Schlafly, E. F., & Finkbeiner, D. P. 2011, ApJ, 737, 103

Schödel, R., Ott, T., Genzel, R., et al. 2002, Nature, 419, 694

Seigar, M. S., Kennefick, D., Kennefick, J., & Lacy, C. H. S. 2008, ApJL, 678, L93

Sellwood, J. A., & Carlberg, R. G. 1984, ApJ, 282, 61

—. 2019, arXiv e-prints, arXiv:1906.04191

Sérsic, J. L. 1963, Boletin de la Asociacion Argentina de Astronomia La Plata Argentina, 6, 41

Shankar, F. 2009, New Astronomy Reviews, 53, 57

Shankar, F., Salucci, P., Granato, G. L., De Zotti, G., & Danese, L. 2004, MNRAS, 354, 1020

Shu, F. H. 2016, ARA&A, 54, 667

Tully, R. B., & Fisher, J. R. 1977, A&A, 54, 661

Tundo, E., Bernardi, M., Hyde, J. B., Sheth, R. K., & Pizzella, A. 2007, ApJ, 663, 53

Vika, M., Driver, S. P., Graham, A. W., & Liske, J. 2009, MNRAS, 400, 1451

Wickham, H. 2009, ggplot2: Elegant Graphics for Data Analysis (Springer-Verlag New York). http://ggplot2.org

Xu, W., & Goodman, J. 2018, MNRAS, 480, 4327